

Phase 3: why keyword scoping fails

measured on GR10, task = "design integration tests for user registration"

v1 · match task keywords against file names

cues: registr... · utente

selected — 3 of 9

GUIRegistrazioneUtente
ControllerRegistrazioneUtente
Utente

MISSED — 6 of 9

ControllerServiziSistema *the facade*
ServizioPersistenza *persistence*
ServizioUtenti *domain service*
GUIHomePage
Cittadino · OperatoreComunale

the pattern, across all 8 views

persistence boundary missed 8/8

system facade missed 7/8

dashboard boundary missed 6/8

these are named for their **architectural role**,
never for the task — so a task keyword
can never reach them

v2 · seed from three sources, then expand

repository graph
40 types · 97 edges · cached

task seeds
one query per
required_information item

centrality seeds
top-k hub nodes

layer-coverage seeds
top-n per package
catches thin-but-critical layers

expand 1 hop · union

cut only for token budget
fits → scope_strategy = "full"

centrality alone is not enough

it does catch the facade: ControllerServiziSistema
ranks #2 of 40 (degree 14 vs 4 for the next Control)

but ServizioPersistenza ranks only #16 (degree 3):
everything reaches persistence, yet only through
two services. top-12 centrality still misses it.

layer coverage catches it. top-2 per package =
8 seeds, and recovers the facade, the dashboard
and the persistence boundary together.

scoping strategy	recall	views fully covered	files selected
file-name keywords (v1)	42%	0 of 8	9.1 / 40
+ full-text grep	89%	1 of 8	28.9 / 40
name + 1 hop	95%	5 of 8	24.0 / 40
content + 1 hop (v2)	100%	8 of 8	37.9 / 40

and note the last column

full coverage needs 38 of 40 files.
at 72k tokens the whole repo already fits —
so on this data, scoping that works is
barely scoping at all.

keep "full" as a comparison arm.